ROVERSETTER

Mechanical Stability & Moving-Load Investigation

Three historical physical prototypes · Retrospective analysis · Uncompleted classroom validation

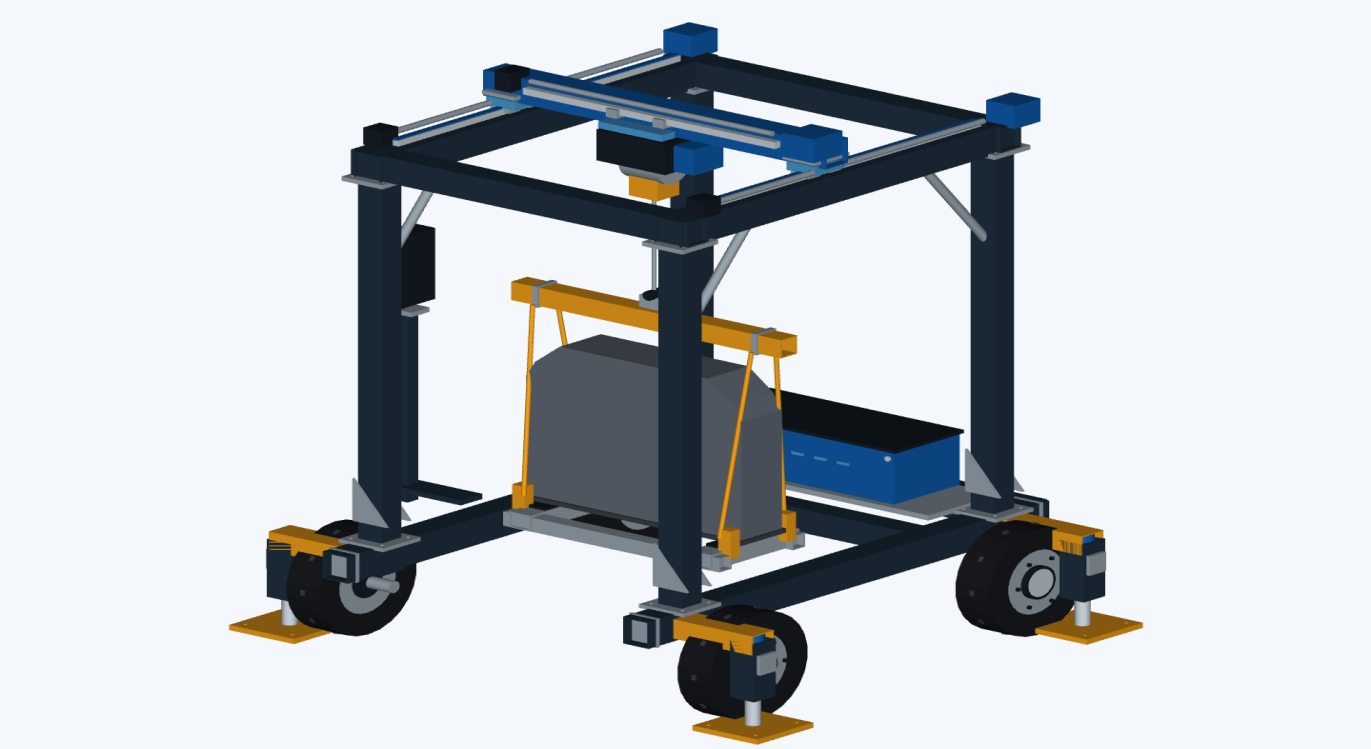


Figure 1. Current saved/exported CAD view of the designer’s original RoverSetter design. The physical prototypes are no longer available; this image is a CAD view, not a test photograph or load-capacity verification.

ABSTRACT

RoverSetter originated in the owner’s monument business after a difficult installation exposed handling limitations. Three physical builds were attempted. In the third integrated test, an approximately 70-pound load hung beyond the front wheel line; both rear wheels lifted nearly together, and forward pitching became more pronounced when the motor-powered chassis began accelerating. This retrospective investigation evaluates a simplified front-pivot static model and shows why greater forward load travel reduces calculated rear support. The historical root cause, especially the motor-startup transient, remains unresolved.

RESEARCH INTEGRITY

Design history: the designer reports developing and building three physical RoverSetter prototypes and maintaining CAD models throughout the project. The current files are updated saves and exports of those models. Analysis: reproduced calculations, model code and diagrams. Pending: new measurements and independent technical review.

This report provides educational analysis, not a design, construction, operating, or load-rating specification for heavy handling machinery.

01 / Historical engineering record

THE PROBLEM AND THE THREE BUILDS

A monument installation using difficult-to-handle equipment took roughly three to four hours and damaged a stone the business owner then replaced. RoverSetter was developed to explore more controllable monument positioning on cemetery terrain. The goal of handling up to approximately 2,000 lb was aspirational, never demonstrated or certified.

Iteration	Presently supportable history	Missing records
Prototype 1	Initial mechanical proof-of-concept; user reports physical construction.	Component designs, dates, test measurements
Prototype 2	Mechanical revision because earlier mechanisms were unsatisfactory; user reports physical construction.	Which features changed, quantitative improvements
Prototype 3	Integrated hoist/positioning and motorized motion; reported approximate 70-lb test with pitching/rocking.	As-built geometry, photographs, motor data, force measurements

LATEST RECALLED TEST SEQUENCE

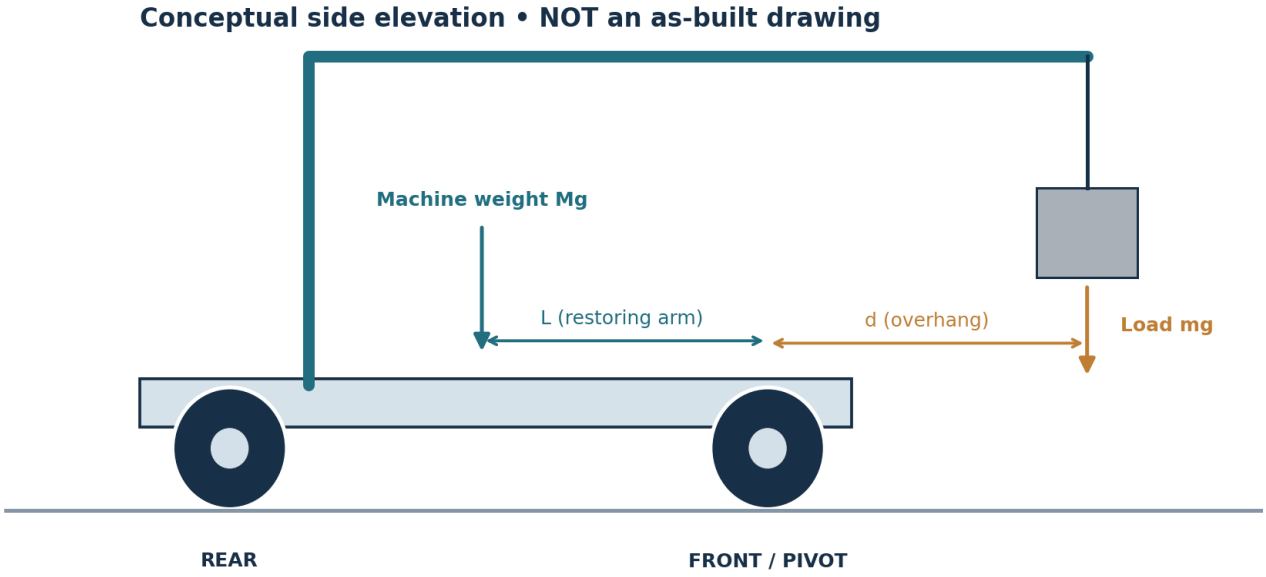
Time / phase	Owner’s observation	Interpretation status
At rest	Hard surface; wheels only; ~70 lb suspended ahead of front support; both rear wheels rose nearly together and rocked/settled.	Supports pitch-pattern interpretation, not proof of tipping threshold.
Motor startup	Powered chassis began accelerating forward; pitching became immediately more pronounced.	Observed timing, not a measured explanation of startup forces.
Moving-load intention	Stone intended to travel relative to chassis via positioning mechanism.	Exact actual travel and timing during this test unmeasured.

CORRECTION LOG

An earlier description of two wheels on one lateral side lifting was corrected by the user to both rear wheels rising nearly together. These are evolving recollections of one test, not two independently documented events.

Evidence: project-owner recollections and October 2026 corrections. The project owner checked the CAD geometry against the original design; original as-built measurements, machine mass properties and time-resolved test data are unavailable.

02 / Mechanical principle and free-body model



IDEALIZED LONGITUDINAL STATIC EQUILIBRIUM

Take moments about the front wheel-contact line. Let M be machine mass, L the machine center-of-mass restoring arm behind the front wheels, m the supported load mass, d the forward overhang of the vertical suspension line, and B the distance between front and rear wheel-contact lines.

$$R = M\,g\,L - m\,g\,d \;;\; N_{rear} = R / B$$

$$\lambda = m\,d / (M\,L) \;;\; d_{crit} = M\,L / m$$

When λ is below 1, the assumed all-wheel equilibrium calculation gives positive total rear reaction. At λ = 1, the ideal rear reaction is zero. Above 1, the unconstrained model requires an unphysical negative rear contact force: the original assumed static contact configuration is infeasible.

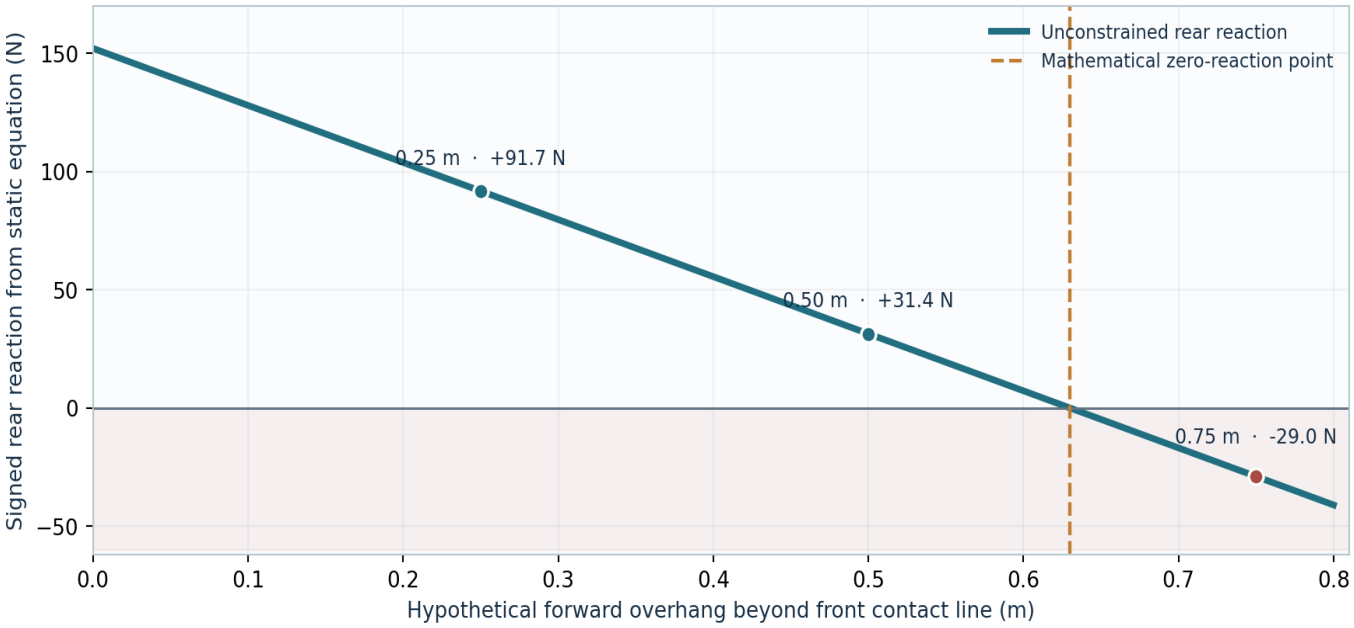
WHAT THIS MODEL CAN AND CANNOT DO

This is a gravity-only, rigid-body, stationary, level-ground calculation. It can show how a load placed farther forward changes the net pitch moment and possible rear reaction. It cannot determine the original prototype’s mass, whether any wheels actually lifted at the calculated threshold, how individual wheel loads divided, structural flex, cable forces when swinging, or behavior during motor startup.

The free-body diagram represents the stated idealized model, not a dimensional machine survey. A negative calculated reaction is not a negative measured force and is not a real support pulling down on a wheel.

03 / Reproducible numerical investigation

Forward overhang lowers theoretical rear support



MODEL ONLY: M=40 kg, L=0.50 m, B=1.29 m (assumed); m≈31.75 kg (recalled estimate). Negative values are INFEASIBLE contacts.

ONE EXPLICITLY HYPOTHETICAL BASELINE

The illustration uses M = 40 kg and L = 0.50 m as hypothetical machine mass properties; B = 1.29 m comes from the designer’s CAD wheel spacing, not an independently measured as-built value; m ≈ 31.75 kg converts the user-recalled 70-lb test stone. No numerical scenario is a recovered physical test.

Assumed d	Ratio λ	Signed rear reaction	Mathematical result
0.25 m	0.397	+91.7 N	Positive equilibrium rear reaction
0.50 m	0.794	+31.4 N	Positive equilibrium rear reaction
0.75 m	1.191	−29.0 N	Assumed contact equilibrium infeasible

SENSITIVITY IS THE GENERALIZABLE RESULT

$$dN_{rear} / dd = -m\,g / B$$

For fixed machine geometry and load mass, predicted rear reaction decreases linearly as forward overhang increases. The modeled threshold around 0.630 m for these assumed parameters is a mathematical boundary only—not a tested machine travel limit or design requirement.

Verification: the included model reproduces the three v11 examples, checks vertical force and pitch moment balance, and passes seven automated tests. These tests verify code consistency, not physical truth.

04 / Moving carriage versus motorized chassis

WHY POSITIONING TRAVEL MATTERS

The machine was intended to move the stone with its positioning mechanism. A load that translates forward relative to the wheel supports increases overhang even if its mass remains unchanged. A moving carriage may also shift part of the machine mass.

ORIGINAL DESIGN-ANALYSIS OMISSION / OWNER RECOLLECTION

The owner recalls treating the suspended stone as fixed at one position in the original stability calculations, despite designing a mechanism that could advance it. This omitted a stability check over the positioning range. The original calculations or notebook have not been recovered; this is a retrospective account of the assumption, not proof that the omission alone caused the observed failure.

$$R(x) = g\,[\,M_0 L_0 + M_c (L_{c0} - x) - m (d_0 + x)\,]$$

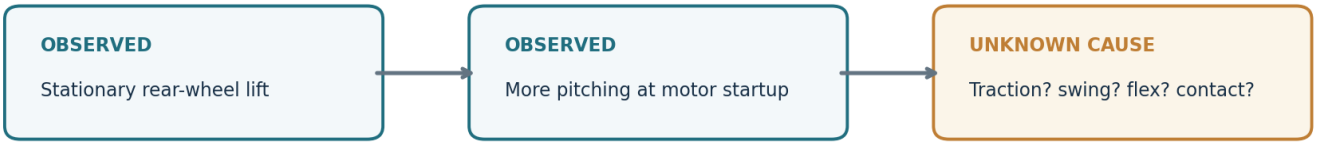
Here x is forward travel relative to the chassis, M_0 and L_0 describe stationary chassis components, M_c and L_{c0} describe carriage mass and initial restoring arm, and d_0 is the initial suspended-load overhang. This extension assumes that the carriage and supported load really translate by the same x and remain quasi-static.

$$dN_{rear} / dx = -(m + M_c)\, g / B$$

CHASSIS TRANSLATION IS A DIFFERENT PHYSICAL PROCESS

Driving the entire chassis forward at steady speed, without changing positions relative to the wheels, does not change the gravity-only arms. Motor startup introduces traction, drive torques, inertia, changing wheel contact, and possible suspended-load motion. Our historical motor acceleration and load travel were not measured, so a unique dynamic cause cannot be inferred.

The startup observation is evidence; its cause is not yet identified



CRITICAL MODEL DISCREPANCY

Ordinary forward acceleration of an idealized rigid vehicle on level ground generally shifts support load toward the rear, whereas the historical recollection says the forward pitch worsened at motor startup. This discrepancy argues against asserting a simplistic acceleration-caused forward tip. It instead highlights missing coupled mechanical effects and absent time-resolved data.

05 / Engineering trade-offs and boundaries

A FOCUSED COMPARATIVE RESULT, NOT A REDESIGN

Changing the hypothetical load position produces a calculable change in rear reaction. For the v11 illustrative values, a 0.10 m difference in overhang corresponds to about 24.1 N difference in the unconstrained total rear reaction. Separately, a hypothetical 0.10 m difference in the machine restoring arm corresponds to about 30.4 N, all else fixed.

Comparison variable	Predicted effect (ideal model)	Trade-off or unresolved concern
Relative forward travel	More overhang → lower rear reaction.	Positioning range vs static support margin; cannot determine practical limit.
Restoring-arm location	More restoring arm → higher rear reaction.	Machine geometry and overall system distribution would change.
Carriage mass movement	Carriage can contribute additional load transfer.	Actual moving mass not recovered from original prototype.
Motor startup	Static model does not predict transient behavior.	Requires force, geometry, motion, and contact data.

EXPLICITLY UNRESOLVED PHYSICAL FACTORS

Unknown original M, L, B, d; stone relative travel; carriage mass; contact patch locations; pavement grade; motor drive axle/torques; acceleration; hoist cable angle; load sway; structural stiffness; local joint compliance; transient wheel contact forces.

CONCLUSIONS THAT CANNOT BE CLAIMED

The original failure was not conclusively solved. A real machine was not structurally verified, load-tested or redesigned to a proven safe specification. The current CAD file documents the designer’s model, but alone cannot independently corroborate the historical prototype builds or tests. A simulated threshold is not a safe machine working limit. No external adoption or experimental validation has occurred within this research package.

Engineering ethics: this package intentionally contains no procedure for building, uprating, or testing heavy-load monument machinery.

06 / Experimental validation protocol

PLANNED ONLY — NO EXPERIMENTS OR INDEPENDENT MEASUREMENTS INCLUDED

THE TESTABLE QUESTION

Does forward repositioning of a very small, securely attached mass on a lightweight, stationary tabletop object decrease the measured total rear-support reaction under teacher-approved, stable conditions? This is a safe analogy for part of the static model; it does not recreate the full-scale failure or motor-startup dynamics.

Item	Teacher-supervised classroom plan
Scope and safety	Harmless low-profile attached mass; four supports; level tabletop; fully supported and stable throughout; no hoist, suspended weight, motors, stone, or deliberate tipping.
Pre-register	Record apparatus mass, rear/front spacing, position reference, mass location, sensor units/resolution, and expected negative slope.
Measurements	Use approved low-force/contact measurement equipment if available; repeat 5 times at each of a few approved, stable mass positions.
Analysis	Calculate total rear force, per-position mean and sample SD, and measured-versus-predicted slope with appropriate units.
Review	Show teacher the raw logs and calculations; document critique and corrections.

STANDARDS OF EVIDENCE

The workbook contains a blank trial log and an analysis script that refuses to analyze missing data. Completion requires real observations, recorded measurement uncertainty, independently checked mathematics, and external teacher review. A numerical prediction—even one perfectly matched to its own code—is not experimental validation.

The full one-page proposal and trial CSV are included separately for teacher approval. The teacher decides whether equipment, conditions, and supervision are appropriate.

07 / Deliverables, provenance & next steps

WHAT HAS BEEN COMPLETED

Deliverable	Status and evidence
Historical owner account	Three physical builds and Prototype 3 test recollections recorded, not independently corroborated
Design CAD	239-solid current export of the original design model; design dimensions designer-confirmed; no surviving as-built survey
Static model and sensitivities	Python source; three v11 reference scenarios; 7 passing automated tests
Moving-load extension	Symbolic and coded quasi-static formula; no original carriage measurements
Classroom protocol	Prepared and safety-constrained; no trials conducted
Workbook and data log	Editable formulas and blank real-data log
Independent external technical review	Not yet completed

NEXT EVIDENCE REQUIRED

1. Seek dated Prototype 3 photos or notes that corroborate the owner-checked design dimensions. 2. Ask a teacher to review the harmless tabletop protocol. 3. Record raw classroom data if approved, without retroactively inventing measurements. 4. Compare observations with predictions and obtain documented external critique. 5. Update conclusions only after that evidence exists.

ATTRIBUTION AND SOURCES

Project history: the designer reports maintaining the original RoverSetter CAD models and exporting updated versions; the older folder was deleted during file organization. Engineering basis: longitudinal statics and reproducible numerical models. Dates: current STEP export timestamps describe the export, not the initial design date. Evidence limits: original prototype photos, direct wheel-force records and independently measured as-built geometry are unavailable.

FINAL SCIENTIFIC CONCLUSION

The reported forward hoist load and nearly symmetric rear-wheel lift are consistent with a forward-pitch instability mechanism. The mathematical model proves only how a hypothetical static rear reaction varies with overhang and mass distribution. The historical trigger and increased pitching during motor startup remain unsolved. What the investigation demonstrates is a systems-engineering lesson: successful local mechanisms do not guarantee stable behavior of a complete moving system.